

Companion LVIII: The Weak-Lensing Bispectrum in the Relaxation-Driven Cyclic Cosmology

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Abstract

The weak-lensing bispectrum probes the non-Gaussian structure of the gravitational potential by measuring the three-point correlations of the cosmic shear field. In the standard cosmological model, the bispectrum arises from nonlinear structure formation and the mode coupling induced by gravitational evolution. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the nonlinear coupling, the potential depth, and the three-point structure of the shear field in a characteristic way. This Companion develops a continuous description of the weak-lensing bispectrum in RDCC, deriving how the relaxation scale influences the bispectrum amplitude, the configuration dependence, and the observational signatures in cosmic shear surveys. The resulting framework provides a distinctive probe of the RDCC gravitational field beyond the two-point level.

1 Introduction

Weak gravitational lensing probes the integrated gravitational potential along the line of sight. While the power spectrum captures the Gaussian component of the shear field, the bispectrum captures the non-Gaussian structure induced by nonlinear gravitational evolution. The bispectrum is sensitive to the mode coupling between large and small scales and provides a powerful probe of the gravitational field.

In the standard cosmological model, the bispectrum arises from nonlinear structure formation and the coupling of density modes. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the nonlinear coupling, the potential depth, and the three-point structure of the shear field in a scale-dependent manner, producing a distinctive signature in the weak-lensing bispectrum.

2 The RDCC Nonlinear Potential

The nonlinear gravitational potential evolves as

$$\Phi_{\text{NL,RDCC}}(k, a) = \Phi_{\text{NL}}(k, a) \left[1 - \chi_{\text{NL}} \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (1)$$

The nonlinear mode-coupling kernel becomes

$$F_{2,\text{RDCC}}(\mathbf{k}_1, \mathbf{k}_2) = F_2(\mathbf{k}_1, \mathbf{k}_2) \left[1 - \chi_F \left(\frac{k_1 + k_2}{2k_{\text{rel}}} \right)^\alpha \right]. \quad (2)$$

This modification reflects the scale-dependent evolution of the gravitational potential.

3 The Weak-Lensing Bispectrum

The convergence bispectrum is defined as

$$B_\kappa(\ell_1, \ell_2, \ell_3) = \int_0^{\chi_H} d\chi \frac{W^3(\chi)}{\chi^4} B_m\left(\frac{\ell_1}{\chi}, \frac{\ell_2}{\chi}, \frac{\ell_3}{\chi}; z(\chi)\right). \quad (3)$$

In RDCC, the matter bispectrum becomes

$$B_{m,\text{RDCC}} = B_m \left[1 - \chi_B \left(\frac{k_1 + k_2 + k_3}{3k_{\text{rel}}} \right)^\alpha \right]. \quad (4)$$

The convergence bispectrum becomes

$$B_{\kappa,\text{RDCC}} = B_\kappa \left[1 - \chi_\kappa \left(\frac{\ell_1 + \ell_2 + \ell_3}{3\ell_{\text{rel}}} \right)^\alpha \right]. \quad (5)$$

4 Configuration Dependence and RDCC Signatures

The relaxation mechanism enhances the coherence of large-scale modes. The weak-lensing bispectrum therefore exhibits:

- enhanced equilateral configurations at large angular scales,
- suppressed squeezed configurations at small scales,
- a characteristic turnover near ℓ_{rel} ,
- a modified redshift dependence of the bispectrum amplitude.

These signatures provide a direct observational probe of the RDCC gravitational field beyond the two-point level.

5 Redshift Evolution of the RDCC Bispectrum

The redshift evolution of the RDCC bispectrum is given by

$$\frac{\partial B_{\kappa,\text{RDCC}}}{\partial z} = \frac{\partial B_\kappa}{\partial z} \left[1 - \chi_\kappa \left(\frac{\ell_1 + \ell_2 + \ell_3}{3\ell_{\text{rel}}} \right)^\alpha \right] - B_\kappa \chi_\kappa \alpha \left(\frac{\ell_1 + \ell_2 + \ell_3}{3\ell_{\text{rel}}} \right)^\alpha \frac{\partial \ln \ell_{\text{rel}}}{\partial z}. \quad (6)$$

This term is unique to RDCC and provides a direct probe of the relaxation scale.

6 Conclusion

The weak-lensing bispectrum in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential and the nonlinear matter distribution. The relaxation mechanism modifies the nonlinear coupling, the potential depth, and the three-point structure of the shear field in a scale-dependent manner, producing distinctive signatures in cosmic shear surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of non-Gaussianity, matter evolution, and the interplay between light and gravity in RDCC.

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